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ROLL NO : 23I-0058

SECTION : BSE – 6A

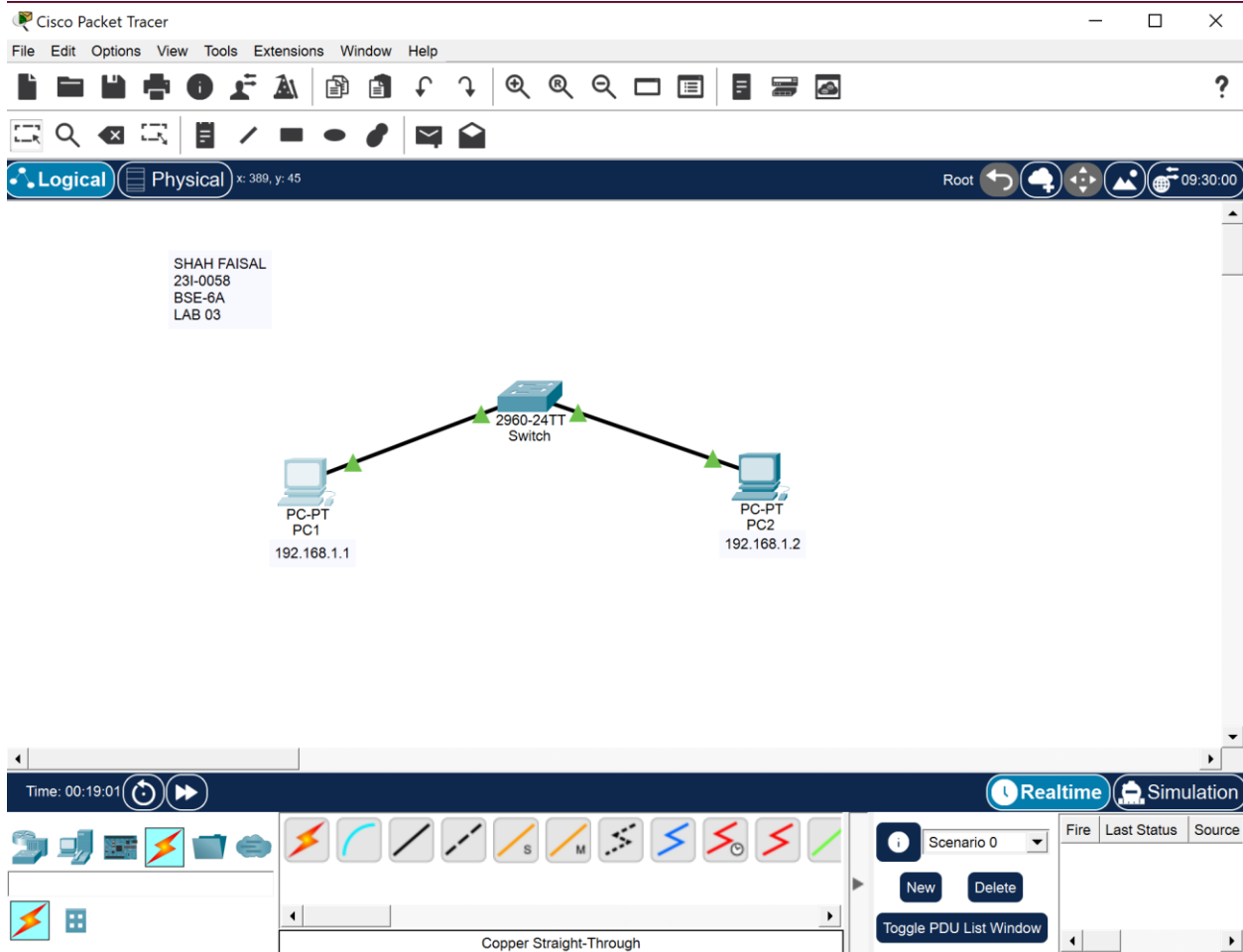
COMPUTER NETWORK LAB

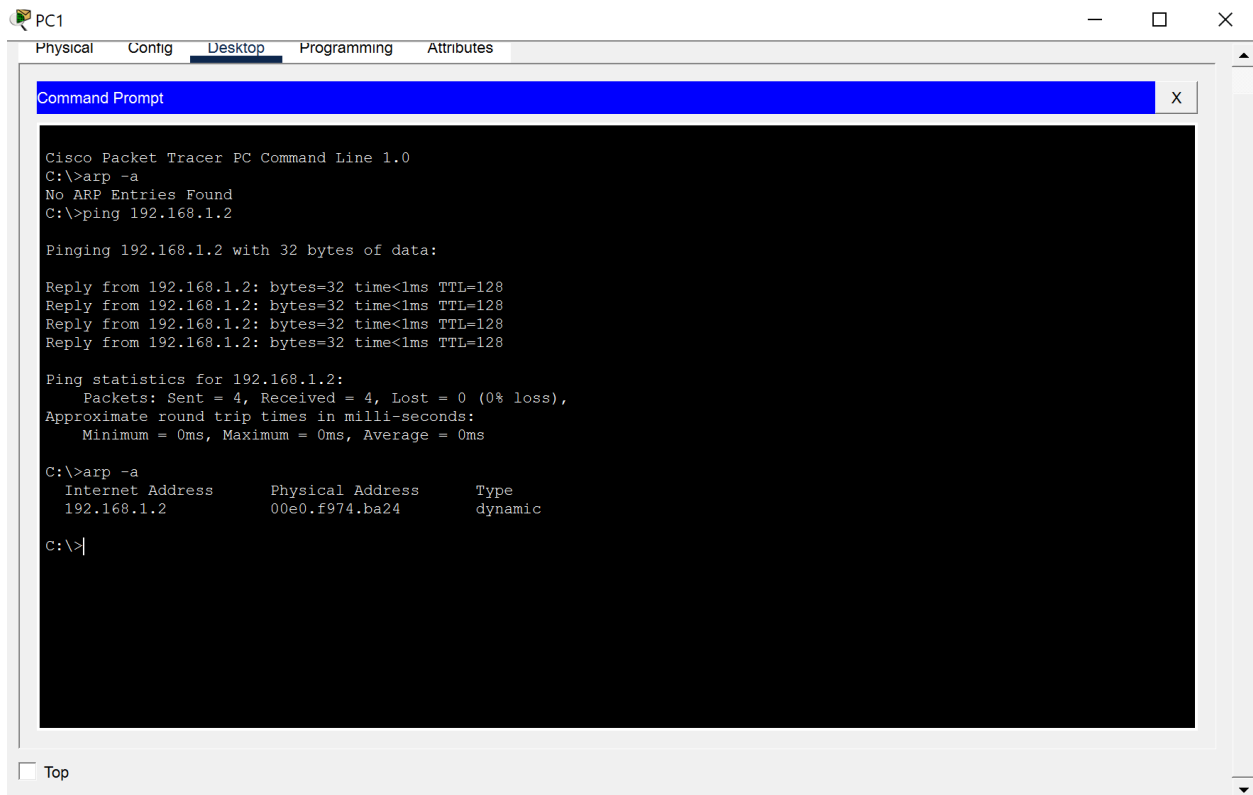
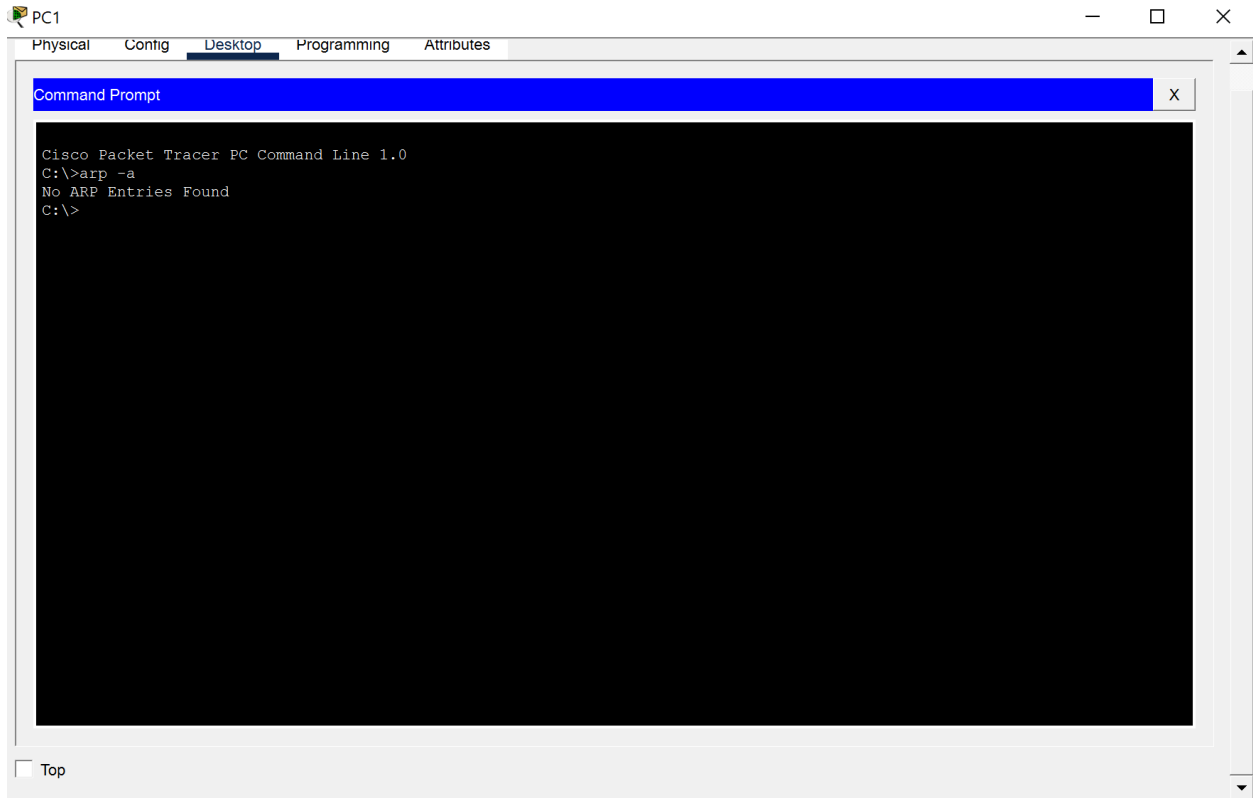
BSSE SPRING 2026

LAB 03

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: Simulating ARP Operation (5 Marks)





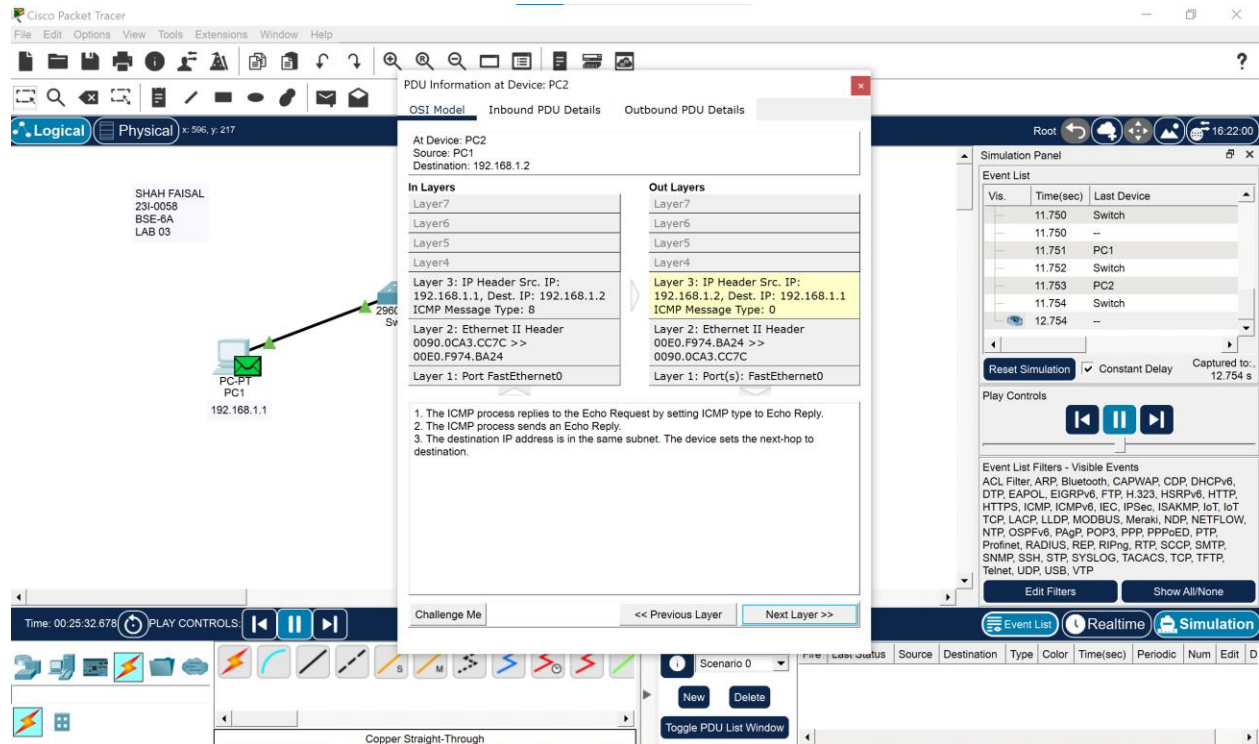
ARP helps devices on the same network find each other's hardware (MAC) addresses using their IP addresses.

When PC1 wants to send data to PC2, it first looks in its ARP table. If PC2's MAC address isn't there, PC1 sends a broadcast message asking, "Who has 192.168.1.2?"

PC2 replies with its MAC address, and PC1 saves it in the ARP table.

Now, future communication is faster because PC1 already knows PC2's MAC address.

Task 2: ICMP Simulation (5 Marks)



The screenshot displays the Cisco Packet Tracer interface during an ICMP Echo simulation. The main workspace shows a network topology with PC1 (192.168.1.1) and PC2 (192.168.1.2) connected via a switch. The 'PDU Information at Device: PC2' window is open, showing the details of the received packet.

PDU Information at Device: PC2

OSI Model	Inbound PDU Details	Outbound PDU Details
Layer 7	Layer 7	Layer 7
Layer 6	Layer 6	Layer 6
Layer 5	Layer 5	Layer 5
Layer 4	Layer 4	Layer 4
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.2 ICMP Message Type: 8	Layer 3: IP Header Src. IP: 192.168.1.2, Dest. IP: 192.168.1.1 ICMP Message Type: 0	Layer 3: IP Header Src. IP: 192.168.1.2, Dest. IP: 192.168.1.1 ICMP Message Type: 0
Layer 2: Ethernet II Header 0090.0CA3.CC7C >> 00E0.F974.BA24	Layer 2: Ethernet II Header 00E0.F974.BA24 >> 0090.0CA3.CC7C	Layer 2: Ethernet II Header 00E0.F974.BA24 >> 0090.0CA3.CC7C
Layer 1: Port FastEthernet0	Layer 1: Port(s): FastEthernet0	Layer 1: Port(s): FastEthernet0

Simulation Panel

Vis.	Time(sec)	Last Device
	11.750	Switch
	11.750	--
	11.751	PC1
	11.752	Switch
	11.753	PC2
	11.754	Switch
	12.754	--

Event List Filters - Visible Events

ACL Filter, ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

PDU Information at Device: PC2



OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: PC2
 Source: PC1
 Destination: 192.168.1.2

In Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.2 ICMP Message Type: 8
Layer 2: Ethernet II Header 0090.0CA3.CC7C >> 00E0.F974.BA24
Layer 1: Port FastEthernet0

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.2, Dest. IP: 192.168.1.1 ICMP Message Type: 0
Layer 2: Ethernet II Header 00E0.F974.BA24 >> 0090.0CA3.CC7C
Layer 1: Port(s): FastEthernet0



1. The packet's destination IP address matches the device's IP address or the broadcast address. The device de-encapsulates the packet.
2. The packet is an ICMP packet. The ICMP process processes it.
3. The ICMP process received an Echo Request message.

Challenge Me

<< Previous Layer

Next Layer >>

PDU Information at Device: PC2

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: PC2
 Source: PC1
 Destination: 192.168.1.2

In Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.2 ICMP Message Type: 8
Layer 2: Ethernet II Header 0090.0CA3.CC7C >> 00E0.F974.BA24
Layer 1: Port FastEthernet0

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.2, Dest. IP: 192.168.1.1 ICMP Message Type: 0
Layer 2: Ethernet II Header 00E0.F974.BA24 >> 0090.0CA3.CC7C
Layer 1: Port(s): FastEthernet0

1. The ICMP process replies to the Echo Request by setting ICMP type to Echo Reply.
2. The ICMP process sends an Echo Reply.
3. The destination IP address is in the same subnet. The device sets the next-hop to destination.

Challenge Me

<< Previous Layer

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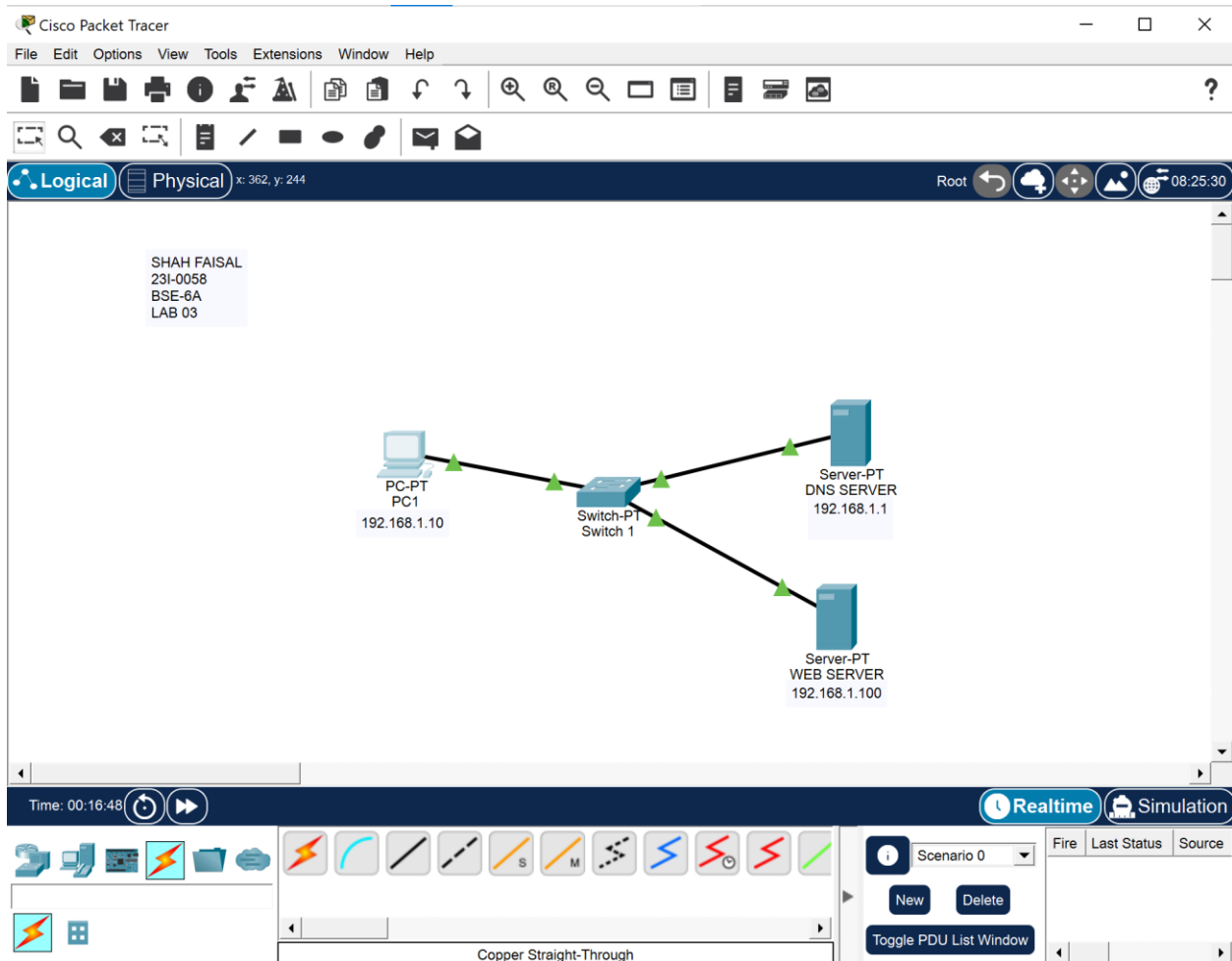
ICMP Echo Request (Type 8):

Sent by the source to check if a destination is reachable. It asks the destination to reply.

ICMP Echo Reply (Type 0):

Sent by the destination in response to an Echo Request. It confirms reachability and network latency.

Task 3: HTTP Encapsulation (5 Marks)



Physical Config **Services** Desktop Programming Attributes

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Address

Add Save Remove

No.	Name	Type	Detail
0	lab3-i230058.com	A Record	192.168.1.100

DNS Cache

PC1

Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://lab3-i230058.com Go Stop

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

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PC1

Physical Config **Desktop** Programming Attributes

Web Browser X

< > URL http://192.168.1.100 Go Stop

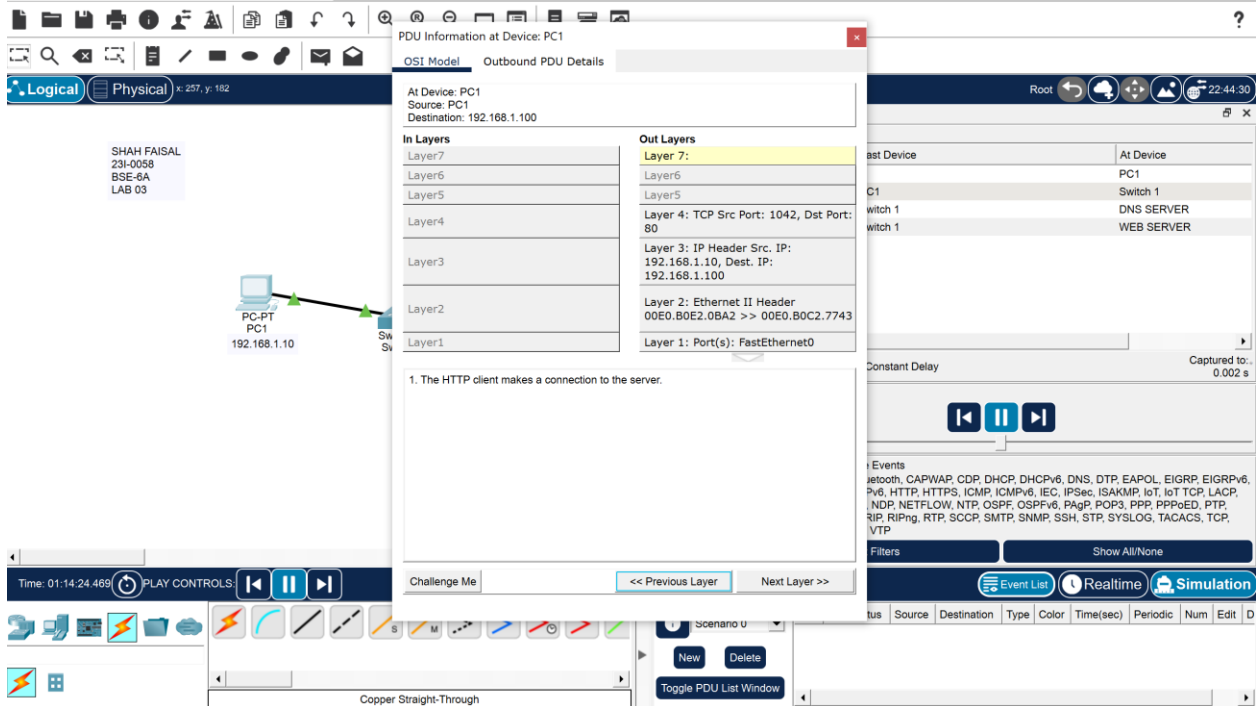
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PDU Information at Device: PC1

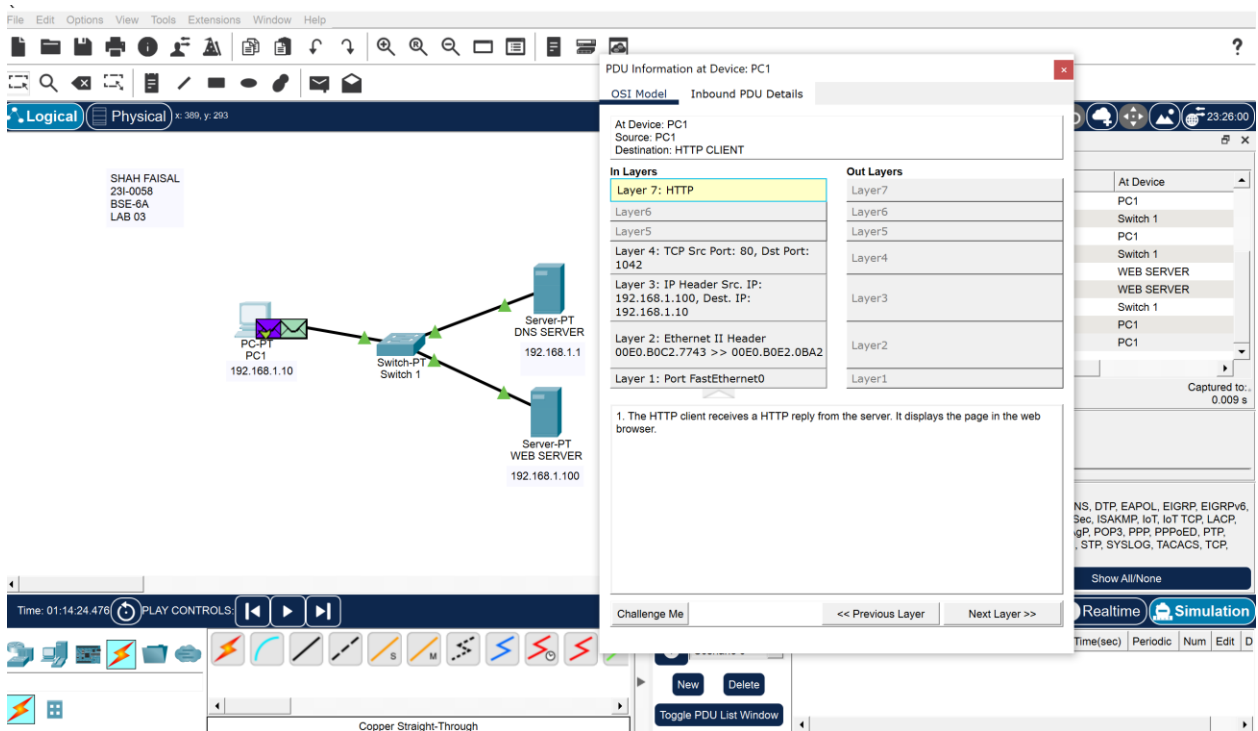
At Device: PC1
Source: PC1
Destination: 192.168.1.100

In Layers	Out Layers
Layer7	Layer 7:
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1042, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.1.100
Layer2	Layer 2: Ethernet II Header 00E0.B0C2.7743 >> 00E0.B0C2.7743
Layer1	Layer 1: Port(s): FastEthernet0

1. The HTTP client makes a connection to the server.

Time: 01:14:24.469 PLAY CONTROLS: [Previous] [Pause] [Next]

Challenge Me << Previous Layer Next Layer >>



PDU Information at Device: PC1

At Device: PC1
Source: PC1
Destination: HTTP CLIENT

In Layers	Out Layers
Layer 7: HTTP	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer2	Layer2
Layer1	Layer1

1. The HTTP client receives a HTTP reply from the server. It displays the page in the web browser.

Time: 01:14:24.476 PLAY CONTROLS: [Previous] [Pause] [Next]

Challenge Me << Previous Layer Next Layer >>

Part 3: HTTP Request Packet

The PC asks the web server for the webpage by sending a GET / request. It says:
"Give me the page for lab3-23I-0058.com".

Part 4: HTTP Reply Packet

The web server sends back the webpage with a 200 OK response. It includes the HTML: <h1>SHAH FAISAL - 23I-0058 - BSE-6A - LAB 03</h1> for the browser to display.